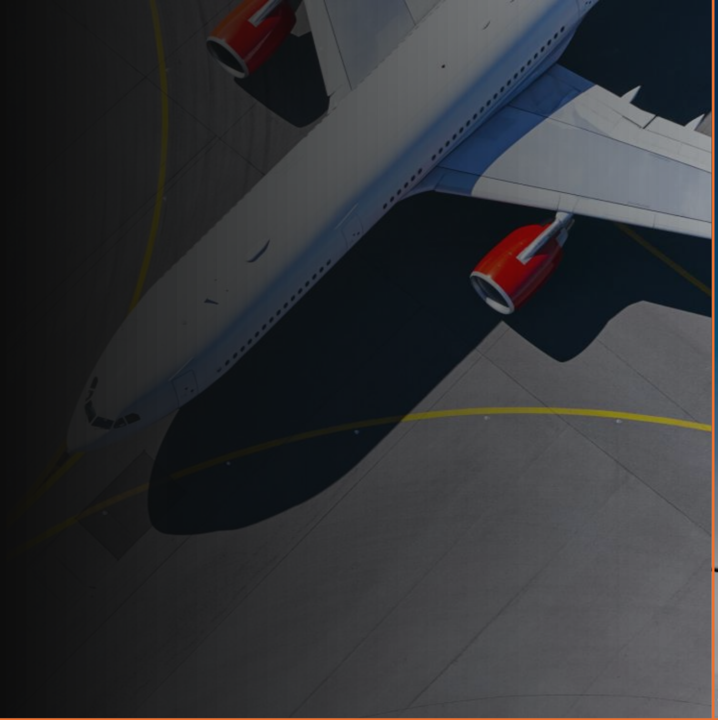
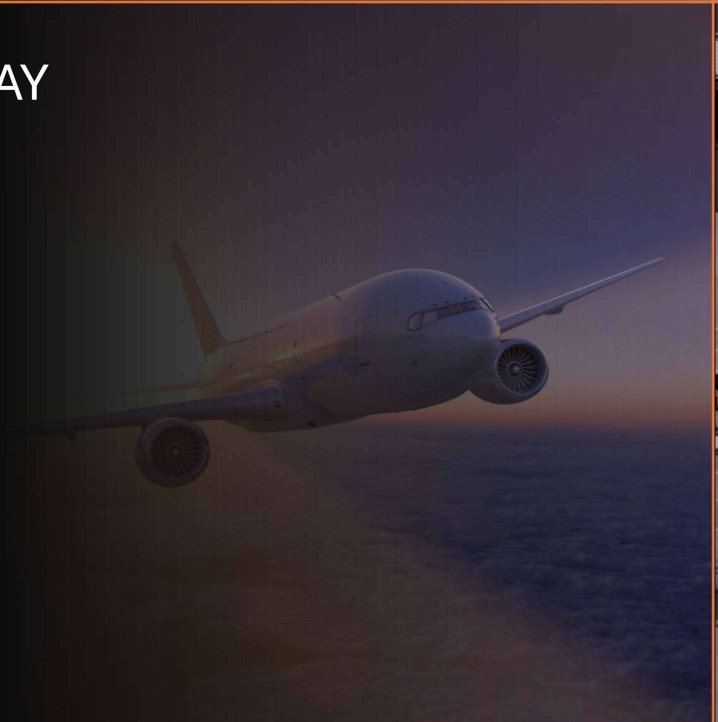
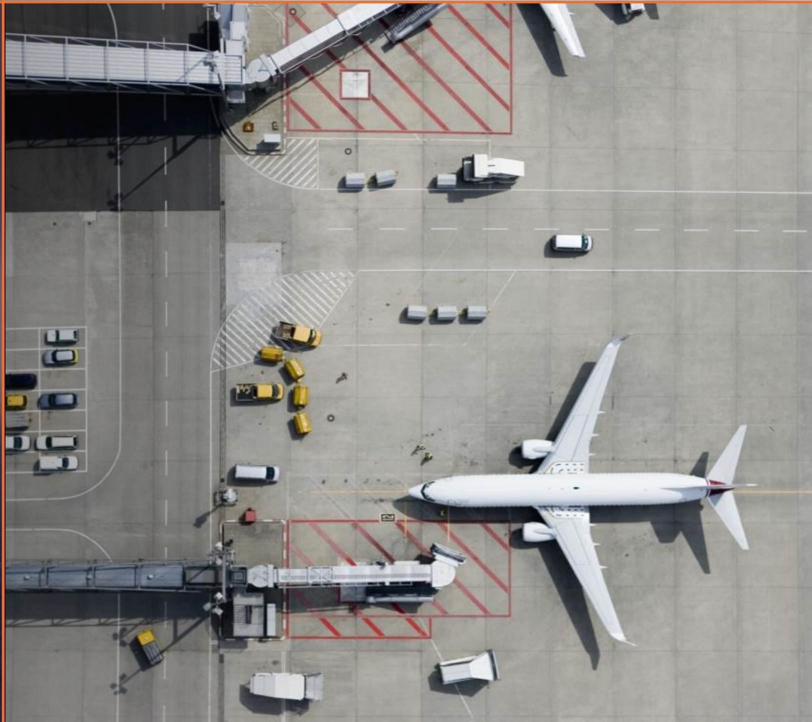


Project 5

AIRLINE OPERATIONS & FLIGHT DELAY ANALYTICS





Project 5: Airline Operations & Flight Delay Analytics

Ocean Star Airlines — Data Analysis & Dashboarding

Sections

1. Scenario
2. Your Role
3. Overview of the Datasets
4. Data Cleaning Section
5. Data Model Slide
6. Exploratory Analysis Section
7. Dashboard Development Section
8. Insight Generation Section
9. Recommendations for Management
10. Executive Summary

Scenario

Ocean Star Airlines operates a busy UK–Europe network with hundreds of flights each week.

Over the last three months, the airline has experienced a combination of operational challenges affecting punctuality and customer satisfaction.

Scenario

Your Task
Use the datasets provided to investigate these issues, identify patterns, and build a dashboard that explains what is happening and what the airline should do next.

2. Airport Bottlenecks
•Major hubs (LHR, AMS, CDG) are experiencing congestion
•Slot restrictions and ATC delays are becoming more frequent
•Ground handling delays are rising during peak hours

4. Technical & Maintenance Issues
•Older aircraft types show repeated technical faults
•Severity levels in maintenance logs are increasing
•Some aircraft are flying with minimal buffer between maintenance cycles

1. Rising Delays Across the Network

- Departure and arrival delays are increasing on key routes
- Short-haul flights are most affected due to tight turnaround windows
- Evening flights show the highest delay growth.

3. Weather-Related Disruptions
•Fog, storms, and low visibility are causing early-morning and winter delays
•Weather patterns vary by region, affecting some airports more than others
•Weather-linked delays are longer and harder to recover from

5. Crew-Related Challenges
•High duty hours are increasing fatigue risk
•Crew shortages are causing last-minute delays
•Fatigue-related delays cluster around early mornings and late evenings

Your Role

You are the Data Analyst for Ocean Star Airlines. Your job is to turn raw operational data into clear, evidence-based insights that management can act on.

What You Are Responsible For

1. Data Cleaning & Preparation

- Fix missing or incorrect times
- Standardise airport codes and formats
- Create calculated fields (delays, turnaround time, risk indicators)
- Merge multiple datasets into a single model

2. Exploratory Analysis

You will investigate:

- Which routes and airports perform worst
- Which delay categories cause the most disruption
- How weather, maintenance, and crew fatigue influence delays
- Patterns across aircraft types and time of day

3. Dashboard Development (Power BI)

You will build a one-page operational dashboard that includes:

- KPI cards
- Trend lines
- Route and airport performance visuals
- Weather, maintenance, and crew impact charts
- A table of the worst 10 flights

4. Insight Generation

You will interpret your findings and answer key questions:

- Why are delays happening?
- Which factors have the biggest impact?
- What should the airline fix first?
- How can performance be improved?

5. Recommendations for Management

Your final output must include:

Clear, evidence-based recommendations

Operational actions the airline can take

Visuals that support your conclusions

A short written summary (150–250 words)

Overview of the 9 Datasets

1. **Flights.csv — Core Operational Data**
Scheduled vs actual times, routes, aircraft types, distances. Forms the backbone of the model and is used to calculate delays, turnaround times, and route performance.

2. **Delays.csv — Root Cause Data**
Weather, Crew, Technical, ATC, Turnaround delay minutes. Helps identify which operational factors contribute most to total delay minutes.

3. **Passengers.csv — Customer & Revenue Data**
Ticket types, fares, baggage, passenger counts. Used to analyse load factors, revenue impact, and customer experience.

4. **Airports.csv — Airport Profiles**
Location, timezone, airport type, hub classification. Supports analysis of airport congestion, regional patterns, and performance differences.

5. **WeatherAPI.csv — Weather Conditions**
Visibility, wind speed, temperature, weather type. Allows investigation of how weather patterns influence delays across airports and seasons.

6. **MaintenanceLogs.csv — Technical Issues**
Severity levels, issues reported, maintenance cycles. Used to link aircraft reliability and technical faults to operational delays.

7. **CrewSchedules.csv — Crew Duty & Fatigue Data**
Duty hours, rest periods, fatigue scores, late reporting, crew shortages. Explains how human factors, such as fatigue and staffing, contribute to delays, especially during early mornings and late evenings.

8. **Bookings.csv — Revenue & Booking Behaviour**
Fares, taxes, booking channels, cancellations, refunds. Used to analyse revenue patterns and customer booking behaviour.

9. **Baggage.csv — Ground Handling & Turnaround Data**
Bag weights, load status, offloads, misroutes. Used to assess weight & balance accuracy, identify causes of mishandled bags, evaluate ground-handling performance, and understand how baggage processes affect turnaround times.

DATA CLEANING & PREPARATION

Why Data Cleaning Matters in Airline Operations

Purpose: Before analysis, the datasets must be aligned, corrected, and merged. Airline data is time-sensitive, operationally critical, and highly interdependent.

Key Issues You Will Fix:

- Missing or incorrect times
- Inconsistent airport codes
- Incorrect date/time formats
- Negative or impossible delays
- Duplicate flight records
- Mismatched aircraft IDs
- Weather and maintenance data not aligned to flight timestamps

TARGET: You will prepare a clean, reliable dataset that accurately reflects real airline operations.

Cleaning the Flights Dataset (Core Operational Data)

Tasks:

- Convert all times to **24-hour format**
- Create **Scheduled Delay** = Actual Departure – Scheduled Departure
- Create **Arrival Delay** = Actual Arrival – Scheduled Arrival
- Fix negative delays (e.g., early arrivals → set to 0 or keep as negative depending on your teaching preference)
- Standardise airport codes to **IATA 3-letter format**
- Remove duplicate Flight IDs
- Check aircraft type consistency (A320, B737, E190 etc.)

Calculated Columns:

- **`DepartureDelayMinutes`**
- **`ArrivalDelayMinutes`**
- **`TurnaroundTime = ActualDeparture – ActualArrival (previous flight)`**
- **`FlightDuration = ActualArrival – ActualDeparture`**

Cleaning the Delays Dataset (Root Cause Data)

Tasks:

- Standardise delay categories: Weather, Crew, Technical, ATC, Turnaround
- Remove blank categories
- Convert delay minutes to numeric
- Group multiple delay entries per flight into a single row

  (Power Query: Group By Sum)

Calculated Columns:

- **`TotalDelayMinutes`**
- **`PrimaryDelayCategory`** (category with highest minutes)
- **`OperationalImpactFlag`** (e.g., >30 mins = High Impact)

Cleaning the Weather Dataset

Tasks:

- Convert weather timestamps to match flight local time
- Standardise weather types (Fog, Storm, Rain, Clear, Snow)
- Fix missing visibility/wind values (use median or nearest-hour reading)
- Remove impossible values (e.g., visibility > 10,000m)

Calculated Columns:

- WeatherRisk = IF(Visibility < 2000 OR WindSpeed > 25, "High", "Normal")
- TemperatureBand = Cold / Mild / Hot

Cleaning the Maintenance Logs

- **Tasks:**
- Standardise severity levels (Low, Medium, High, Critical)
- Remove duplicate maintenance entries
- Align aircraft tail numbers with Flights.csv
- Identify aircraft with repeated faults
- **Calculated Columns:**
- MaintenanceRisk = IF(Severity = "High" OR "Critical", 1, 0)
- DaysSinceLastMaintenance
- FaultFrequency (per aircraft)

Merging Datasets & Building the Data Model

Dataset	Join Key	Relationship
Flights	FlightID	1 → Many with Delays
Flights	AircraftID	1 → Many with MaintenanceLogs
Flights	DepartureAirport / ArrivalAirport	Many → 1 with Airports
Flights	DateTime	Many → 1 with Weather (nearest timestamp)
Flights	FlightID	1 → Many with Passengers
Bookings	FlightID	Many → 1 with Flights
Passengers	BookingReference	Many → 1 with Bookings
Baggage	PassengerID	Many → 1 with Passengers
Baggage	FlightID	Many → 1 with Flights


EXPLORATORY ANALYSIS
(Visual Slide Version)

EXPLORATORY ANALYSIS Overview

What You Will Explore

- Worst performing routes
- Worst performing airports
- Delay categories causing the most disruption
- Impact of weather, maintenance, and crew fatigue
- Patterns by aircraft type
- Patterns by time of day

Purpose

To uncover the operational hotspots driving delays across the network.

Visual Cue

Think of this section as “finding the story” before building the dashboard.

Route Performance Analysis

Key Questions

- Which routes have the highest average delay?
- Are short haul routes more affected than long haul?
- Which city pairs consistently underperform?

Visuals to Use

- Bar Chart → Avg Delay by Route
- Map Visual → Route performance by region
- Scatter Plot → Distance vs Delay

Measures

- AvgDepartureDelay
- AvgArrivalDelay
- FlightsOnTime %

Airport Performance Analysis

Departure Airports

- Which airports cause the longest departure delays?
- Are major hubs (LHR, AMS, CDG) worse than regional airports?

Arrival Airports

- Which airports are most congested?
- Which airports show repeated ground handling delays?

Visuals

- Column Chart → Avg Delay by Departure Airport
- Column Chart → Avg Delay by Arrival Airport
- Heat Map → Airport performance by region

Delay Category Analysis

Categories to Analyse

- Weather
- Crew
- Technical
- ATC
- Turnaround

Key Questions

- Which category causes the most minutes lost?
- Which categories are increasing over time?
- Which categories hit short haul vs long haul differently?

Visuals

- Stacked Bar → Total Delay Minutes by Category
- Line Chart → Category trends over time
- Treemap → Proportion of delay types

Weather Impact Analysis

Key Weather Factors

- Visibility
- Wind speed
- Storms
- Fog
- Temperature

Questions

- How does weather affect delays at different airports?
- Are early morning flights more affected?
- Which weather types cause the longest recovery times?

Visuals

- Bar Chart → Avg Delay by Weather Type
- Scatter → WeatherRisk vs DelayMinutes
- Line Chart → Delays on Fog Days vs Clear Days

Maintenance Impact Analysis

What to Look At

- Aircraft types with repeated faults
- High severity maintenance logs
- Aircraft with minimal buffer between cycles

Questions

- Do older aircraft have more delays?
- Do high severity logs correlate with long delays?
- Are certain aircraft repeatedly causing issues?

Visuals

- Column Chart → Avg Delay by Aircraft Type
- Bar Chart → Delay Minutes by Maintenance Severity
- Table → FaultFrequency per Aircraft

Time of Day & Day of Week Patterns

Time of Day

- Morning fog delays
- Afternoon stabilisation
- Evening congestion

Day of Week

- Friday business travel peaks
- Sunday evening return travel spikes

Visuals

- Line Chart → Avg Delay by Hour of Day
- Column Chart → Avg Delay by Day of Week
- Area Chart → Delay Trend Across the Day

DASHBOARD DEVELOPMENT (POWER BI)

Dashboard Overview

Purpose: To build a one-page operational dashboard that explains where delays are happening, why they are happening, and which factors have the biggest impact.

What Your Dashboard Will Include:

• KPI cards • Delay trend lines • Route performance visuals • Airport performance visuals • Delay category breakdown • Weather, maintenance, and crew impact charts • A table of the worst 10 flights

TARGET: A clear, evidence-based operational dashboard for management.

Recommended Dashboard Layout

Top Row — KPIs

- Avg Departure Delay
- Avg Arrival Delay
- % Flights On Time
- Total Delay Minutes
- High Impact Delay Count (>30 mins)

Middle Row — Trends & Drivers

- Delay Trend (Last 3 Months)
- Delay Minutes by Category
- Delay by Time of Day

Bottom Row — Operational Hotspots

- Route Performance
- Airport Performance
- Worst 10 Flights Table

KPI Cards (Key Metrics)

KPI Cards to Create:

- AvgDepartureDelay
- AvgArrivalDelay
- FlightsOnTime %
- TotalDelayMinutes
- HighImpactDelays (>30 mins)

DAX Measures:

- AvgDepartureDelay = AVERAGE(Flights[DepartureDelayMinutes])
- AvgArrivalDelay = AVERAGE(Flights[ArrivalDelayMinutes])
- FlightsOnTime = DIVIDE(COUNTROWS(OnTimeFlights), COUNTROWS(Flights))
- HighImpactDelays = COUNTROWS(FILTER(Delays, Delays[TotalDelayMinutes] > 30))

Visual Tip:
Use green/amber/red conditional formatting to mimic real airline ops dashboards.

Delay Trend Visual

Visual: Line Chart
X-Axis: Date
Y-Axis: Avg Delay (mins)

What It Shows:

- Whether delays are rising or falling
- Weekly or seasonal patterns
- Impact of weather or maintenance spikes

Teaching Prompt:
“What events might explain the spikes?”

Route Performance Visual

Visuals to Use:

• Bar Chart → Avg Delay by Route
• Map Visual → Route performance by region
• Scatter Plot → Distance vs Delay

Purpose:
Identify the worst-performing routes and compare short-haul vs long-haul.

Airport Performance Visual

Departure Airports
• Which airports cause the longest departure delays?
• Are major hubs worse than regional airports?

Arrival Airports
• Which airports are most congested?
• Which airports show repeated ground handling delays?

Visuals:
• Column Chart → Avg Delay by Departure Airport
• Column Chart → Avg Delay by Arrival Airport
• Heat Map → Airport performance by region

Delay Category Breakdown

Categories:

- Weather
- Crew
- Technical
- ATC
- Turnaround

Visuals:

- Stacked Bar → Total Delay Minutes by Category
- Line Chart → Category trends over time
- Treemap → Proportion of delay types

Purpose:

Shows which operational areas cause the most disruption.

Weather, Maintenance & Crew Impact

Weather Impact

- Avg Delay by Weather Type
- WeatherRisk vs DelayMinutes

Maintenance Impact

- Avg Delay by Severity
- FaultFrequency per Aircraft

Crew Impact

- Avg Delay by FatigueRiskLevel
- CrewShortageFlag Count

Visual Tip:

Use three small multiple visuals for side by side comparison.

Time of Day & Day of Week Patterns

Time of Day

- Morning fog delays
- Afternoon stabilisation
- Evening congestion

Day of Week

- Friday business travel peaks
- Sunday evening return travel spikes

Visuals:

- Line Chart → Avg Delay by Hour of Day
- Column Chart → Avg Delay by Day of Week
- Area Chart → Delay Trend Across the Day

Worst 10 Flights Table

Columns to Include:

- Flight Number
- Route
- Total Delay Minutes
- Primary Delay Category
- WeatherRisk
- MaintenanceRisk
- FatigueRiskLevel

Purpose:
Highlights the flights causing the most operational disruption.

Teaching Prompt:
“Which flights should the airline investigate immediately?”

INSIGHT GENERATION

Insight Generation — Overview

Purpose:
To interpret your dashboard findings and explain **why delays are happening**, **which factors matter most**, and **where the airline should focus its efforts**.

You Will Answer:

• What patterns stand out?
• Which routes and airports are driving delays?
• Which delay categories have the biggest impact?
• How do weather, maintenance, and crew factors influence performance?
• What operational hotspots require immediate attention?

TARGET: Clear, evidence-based insights that management can act on.

Route & Airport Insights

Key Findings to Look For:

• Routes with consistently high departure or arrival delays
• Short-haul routes affected by tight turnaround windows
• Major hubs (LHR, AMS, CDG) showing congestion patterns
• Airports with repeated ground handling or ATC delays
• Routes with high variability (unpredictable performance)

Insight Prompts:

• “Which routes are the biggest contributors to total delay minutes?”
• “Are delays concentrated at specific airports or spread across the network?”
• “Do certain airports cause knock-on delays for later flights?”

Delay Category Insights

What to Identify:

- The top 1–2 categories causing the most total delay minutes
- Categories that are increasing over time
- Categories that disproportionately affect short haul vs long haul
- High impact delays (>30 mins) and their root causes

Insight Prompts:

- “Which delay category is the primary driver of operational disruption?”
- “Are technical or crew delays becoming more frequent?”
- “Which categories should management prioritise first?”

Weather Insights

Patterns to Look For:

- Fog related delays in early mornings
- Storm related delays with long recovery times
- Seasonal spikes (winter vs summer)
- Airports that are more weather sensitive than others

Insight Prompts:

- “Which weather types cause the longest delays?”
- “Do certain airports struggle more during poor visibility?”
- “How does weather impact the rest of the day’s schedule?”

Maintenance Insights

What to Analyse:

- Aircraft types with repeated faults
- High severity maintenance logs linked to long delays
- Aircraft with minimal buffer between maintenance cycles
- Patterns of recurring technical issues

Insight Prompts:

- “Which aircraft types are the least reliable?”
- “Are high severity faults causing cascading delays?”
- “Do certain aircraft need to be rotated out sooner?”

Crew Fatigue & Staffing Insights

Key Patterns:

• High fatigue scores linked to longer delays

• Early morning and late evening flights showing fatigue related disruption

• Crew shortages causing last minute delays

• Late reporting patterns by base or route

Insight Prompts:

• “How does fatigue risk correlate with departure delays?”

• “Are certain bases or routes more affected by crew shortages?”

• “Do last minute swaps contribute to operational instability?”

Time Based Insights

What to Look For:

• Delay peaks by hour of day

• Day of week patterns (Friday peaks, Sunday spikes)

• Evening congestion due to accumulated delays

• Morning delays caused by weather or crew fatigue

Insight Prompts:

• “When do delays peak across the day?”

• “Which days of the week are most problematic?”

• “Are morning delays cascading into evening disruption?”

RECOMMENDATIONS FOR MANAGEMENT

Recommendations for Management — Overview

Purpose:

To provide clear, evidence based operational actions that Ocean Star Airlines can take to reduce delays, improve punctuality, and stabilise network performance.

Your Recommendations Must Be:

- Data driven
- Operationally realistic
- Prioritised
- Linked to dashboard insights
- Actionable within 1–3 months

TARGET: A concise, professional set of recommendations for senior leadership.

1. Improve Turnaround Efficiency on Short-Haul Routes

Why:

Short haul flights show the highest delay growth due to tight turnaround windows and accumulated delays.

Actions:

- Increase ground handling staffing during peak hours
- Introduce a “15 minute readiness check” for high risk routes
- Standardise turnaround SOPs across all bases
- Add buffer time to the worst performing rotations

Expected Impact:

Reduced departure delays and fewer knock on evening disruptions.

2. Strengthen Operations at Congested Airports

Why:

Major hubs (LHR, AMS, CDG) show repeated congestion, ATC delays, and ground handling issues.

Actions:

• Re time flights away from peak congestion windows

• Negotiate improved slot coordination with airport authorities

• Deploy additional ground staff during peak periods

• Prioritise high risk flights for early pushback

Expected Impact:

Smoother departures, fewer ATC related delays, improved on time performance.

3. Address Weather Sensitive Airports and Time Windows

Why:

Fog, storms, and low visibility cause long, unrecoverable delays — especially early mornings.

Actions:

• Add morning buffer time for fog prone airports

• Pre position aircraft the night before during winter months

• Improve weather based decision making (earlier diversions, swaps)

• Use WeatherRisk indicators to prioritise aircraft allocation

Expected Impact:

Reduced morning disruption and fewer cascading delays throughout the day.

4. Enhance Aircraft Reliability & Maintenance Planning

Why:

High severity maintenance logs and repeated faults correlate with long delays.

Actions:

• Rotate high risk aircraft out of tight schedules

• Increase frequency of inspections for older aircraft types

• Introduce predictive maintenance using fault frequency data

• Improve communication between engineering and operations teams

Expected Impact:

Fewer technical delays and improved aircraft availability.

5. Reduce Crew Fatigue & Staffing-Related Delays

Why:

Fatigue scores and crew shortages are strongly linked to early morning and late evening delays.

Actions:

- Increase rest periods for high risk duty patterns
- Add reserve crew during peak disruption windows
- Improve crew reporting processes to reduce late sign ins
- Use FatigueRiskLevel to optimise crew pairing

Expected Impact:

More stable morning departures and fewer last minute crew related delays.

6. Prioritise the Worst 10 Flights for Immediate Intervention

Why:

A small number of flights contribute disproportionately to total delay minutes.

Actions:

- Conduct root cause reviews for each of the worst 10 flights
- Adjust schedules, aircraft allocation, or crew pairing
- Add targeted buffers or operational support
- Monitor these flights daily for 4 weeks

Expected Impact:

Rapid improvement in network performance with minimal cost.

Executive Summary (150–250 words)

Ocean Star Airlines is experiencing rising delays driven by a combination of operational bottlenecks, weather disruption, technical issues, and crew related challenges. Analysis of the last three months shows that short haul routes with tight turnaround windows are the most affected, particularly during evening peaks when accumulated delays reduce schedule resilience. Major hubs such as LHR, AMS, and CDG contribute significantly to departure and arrival delays due to congestion, slot restrictions, and ground handling variability.

Weather related delays, especially fog and storms, create long recovery times and disproportionately impact early morning operations. Technical delays are concentrated among older aircraft types with repeated high severity maintenance logs, indicating the need for improved reliability planning. Crew fatigue and staffing shortages further contribute to morning and late evening disruption, with fatigue scores strongly correlating with departure delays.

To address these issues, the airline should prioritise improving turnaround efficiency, strengthening operations at congested airports, and enhancing maintenance planning. Additional focus on weather sensitive airports, crew fatigue management, and targeted intervention on the worst performing flights will deliver rapid, measurable improvements. These actions will stabilise the network, improve punctuality, and enhance the overall customer experience.